WORKED SOLUTION

STEP III 2021, Question 5

Pure • Polar Curves

STEP III 2021, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Both curves are given as r in terms of θ, so they meet where the two expressions for r agree, and they touch where, in addition, the two values of $\mathrm{d}r/\mathrm{d}\theta$ agree. Writing everything in terms of $c = \cos\theta$ turns the meeting condition into a quadratic in c. A repeated root of that quadratic gives one way of touching, but it isn't the only way: because $\cos\theta$ is itself stationary at $\theta = 0$ and $\theta = \pi$, a simple root at $c = \pm 1$ also gives a double zero in θ.

Solution

Throughout, write $c = \cos\theta$, and

$$f(\theta) = a + 2\cos\theta, \qquad g(\theta) = 2 + \cos 2\theta = 1 + 2\cos^2\theta.$$

Note that $g(\theta) \geq 1$ for all θ, so the second curve never passes through the origin O. Away from O, a point with $r > 0$ has a unique angle θ (up to multiples of 2π). So the curves can only meet at points where they share both r and θ.

Part (i)

The curves meet when $f(\theta) = g(\theta)$, that is $a + 2c = 1 + 2c^2$, or

$$2\cos^2\theta - 2\cos\theta + 1 - a = 0,$$

as required. It helps to name the difference:

$$h(\theta) = f(\theta) - g(\theta) = -(2c^2 - 2c + 1 - a).$$

When do they touch? At a common point, the angle ψ between the radius and the tangent satisfies $\tan\psi = r \big/ \frac{\mathrm{d}r}{\mathrm{d}\theta}$. So two curves through the same point (r, θ) have the same tangent there exactly when they also have the same $\mathrm{d}r/\mathrm{d}\theta$. The curves therefore touch where $h(\theta) = 0$ and $h'(\theta) = 0$. Differentiating,

$$h'(\theta) = -2\sin\theta + 2\sin 2\theta = 2\sin\theta\,(2\cos\theta - 1).$$

So $h'(\theta) = 0$ exactly when $\cos\theta = \frac{1}{2}$ or $\sin\theta = 0$.

Ask yourself The quadratic in c has a repeated root only when $a = \frac{1}{2}$. So why are there other values of a?

A repeated root in c is enough for touching, but it isn't necessary. The map $\theta \mapsto \cos\theta$ is stationary at $\theta = 0$ and $\theta = \pi$. So if $c = \pm 1$ is a root at all, h has a double zero in θ there. The condition $h' = 0$ captures both possibilities.

- $\cos\theta = \frac{1}{2}$: substituting into the meeting equation gives $\frac{1}{2} - 1 + 1 - a = 0$, so $a = \frac{1}{2}$. (Equivalently, the discriminant $4 - 8(1 - a)$ vanishes when $a = \frac{1}{2}$, and then the repeated root is $c = \frac{1}{2}$.) So the curves touch when $a = \frac{1}{2}$, as required.
- $\theta = 0$, $c = 1$: we need $2 - 2 + 1 - a = 0$, so $a = 1$. Then $r = 3$ on both curves.
- $\theta = \pi$, $c = -1$: we need $2 + 2 + 1 - a = 0$, so $a = 5$. Then $r = 5 - 2 = 3$ on both curves, so $r \geq 0$ holds.

Hence the other two values are $a = 1$ and $a = 5$.

Features of the second curve

All three sketches contain $r = 2 + \cos 2\theta$, so we describe it once. It is unchanged by $\theta \to -\theta$ and by $\theta \to \pi - \theta$, so it is symmetric in both axes. Here r goes from a maximum of 3 at $\theta = 0, \pi$ to a minimum of 1 at $\theta = \pm\frac{1}{2}\pi$. It is *not* an ellipse. Near $\theta = \frac{1}{2}\pi$, with $s = \sin\theta$,

$$y = r\sin\theta = (3 - 2s^2)s = 3s - 2s^3, \qquad \frac{\mathrm{d}y}{\mathrm{d}s} = 3 - 6s^2 = -3 \text{ at } s = 1.$$

So as θ moves away from $\frac{1}{2}\pi$ (and s falls below 1), y *increases*. The curve has a smooth indentation (a dimple) at $(0, \pm 1)$, not a cusp, since $\mathrm{d}r/\mathrm{d}\theta = -2\sin 2\theta = 0$ there and the tangent is horizontal. By contrast, $x = r\cos\theta = c + 2c^3$ is increasing in c, so the ends at $(\pm 3, 0)$ are simply rounded.

The first curve $r = a + 2\cos\theta$ is symmetric about the initial line, since it is unchanged by $\theta \to -\theta$. It has $r = a + 2$ at $\theta = 0$, $r = a$ at $\theta = \pm\frac{1}{2}\pi$ and (when $a \geq 2$) $r = a - 2$ at $\theta = \pi$. When $a < 2$, the restriction $r \geq 0$ cuts it off where $\cos\theta = -\frac{1}{2}a$, and it reaches O along those two directions.

Part (ii)

With $a = \frac{1}{2}$,

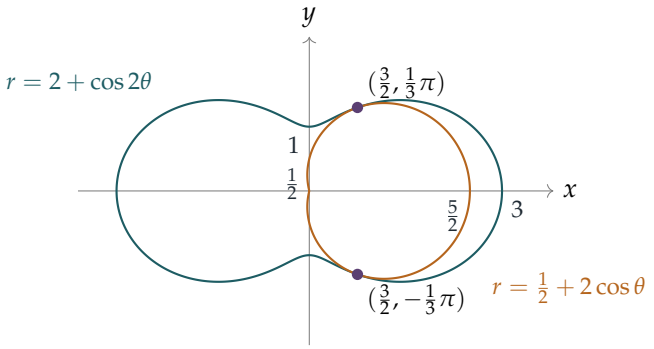
$$h(\theta) = -\left(2c^2 - 2c + \tfrac{1}{2}\right) = -2\left(c - \tfrac{1}{2}\right)^2 \leq 0.$$

So the first curve lies inside the second everywhere and meets it only where $\cos\theta = \frac{1}{2}$. They touch at

$$\theta = \pm\tfrac{1}{3}\pi, \qquad r = \tfrac{1}{2} + 1 = \tfrac{3}{2}.$$

(Check: $2 + \cos\frac{2}{3}\pi = \frac{3}{2}$, and both values of $\mathrm{d}r/\mathrm{d}\theta$ equal $-\sqrt{3}$ at $\theta = \frac{1}{3}\pi$.)

The first curve exists only for $\cos\theta \geq -\frac{1}{4}$, that is $|\theta| \leq \arccos(-\frac{1}{4}) \approx 104.5°$. It runs from $r = \frac{5}{2}$ on the initial line, through $r = \frac{1}{2}$ at $\theta = \pm\frac{1}{2}\pi$, down to O. It arrives at O along the lines $\theta = \pm\arccos(-\frac{1}{4})$, so it has a corner (a pointed tip) at the origin, just to the left of the y-axis.

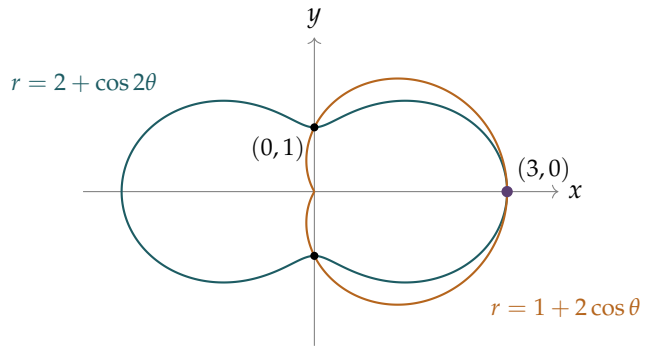


Part (iii)

The case $a = 1$. Here $h(\theta) = -(2c^2 - 2c) = 2c(1 - c)$. The curves meet where $c = 1$ or $c = 0$.

- They touch at $\theta = 0$, $r = 3$, the point $(3, 0)$. Near $\theta = 0$ we have $0 < c < 1$, so $h > 0$ and the first curve is just outside the second on both sides of the point of contact.
- At $\theta = \pm\frac{1}{2}\pi$, $r = 1$ on both curves: the points $(0, \pm 1)$, the bottoms of the dimples. Here h changes sign with c, so the curves *cross*. The first curve is outside for $|\theta| < \frac{1}{2}\pi$ and inside for $|\theta| > \frac{1}{2}\pi$.

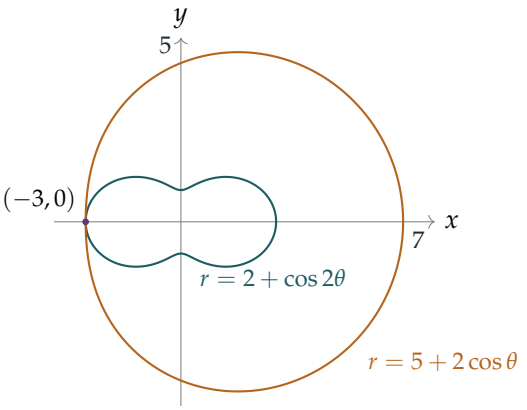
The first curve exists for $\cos\theta \geq -\frac{1}{2}$, i.e. $|\theta| \leq \frac{2}{3}\pi$. It reaches O along the lines $\theta = \pm\frac{2}{3}\pi$, making a corner there.



The case $a = 5$. Here $h(\theta) = -(2c^2 - 2c - 4) = 2(2 - c)(1 + c) \geq 0$, with equality only at $c = -1$. So the first curve lies entirely outside the second and touches it only at

$$\theta = \pi, \qquad r = 3,$$

the point $(-3, 0)$. Since $a > 2$, now $r > 0$ throughout, so the first curve is a closed loop (a limaçon with no dimple, because $a \geq 4$). It has $r = 7$ at $\theta = 0$, $r = 5$ at $\theta = \pm\frac{1}{2}\pi$ and $r = 3$ at $\theta = \pi$.



Checks

- At each touching point, compare both values of r and of $\mathrm{d}r/\mathrm{d}\theta$. For $a = 1$ at $\theta = 0$, both $r = 3$ and both derivatives are 0; the same holds for $a = 5$ at $\theta = \pi$.
- A computer scan over $-3 \leq a < 9$ in steps of 0.01 found $h = h' = 0$ (with $r \geq 0$) only for $a = \frac{1}{2}$, 1 and 5. This matches the factorisation $h' = 2\sin\theta(2\cos\theta - 1)$.
- The point $(0, 1)$ lies on the second curve, and a nearby point at $\theta = \frac{1}{2}\pi + 0.1$ has $y \approx 1.015 > 1$. This confirms the dimple.

Common slips

- Stopping at the discriminant and missing $a = 1$ and $a = 5$. A simple root in $c = \cos\theta$ can still be a double root in θ.
- Drawing $r = 2 + \cos 2\theta$ as an ellipse, or giving it cusps at $(0, \pm 1)$ instead of smooth dimples.
- Drawing the inner loop of the limaçon for $a = \frac{1}{2}$ or $a = 1$. That loop has $r < 0$, which the question excludes.

Beyond STEP

Touching is really a statement about multiplicity. Two curves are tangent where the difference of their equations has a repeated zero along the curve, and the parametrisation can create or hide such zeros, as $\cos\theta$ did here. In a first course on curves, this becomes the idea of the order of contact, and the curvature $\left(r^2 + 2r'^2 - rr''\right) / \left(r^2 + r'^2\right)^{3/2}$ of a polar curve explains exactly where the dimples of $2 + \cos 2\theta$ come from. For more on

parametrised curves and their geometry, try Siklos, *Advanced Problems in Mathematics*, and for the analysis behind smoothness and tangency, Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.